# Lecture 4

*Externalities*

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September 5, 2025

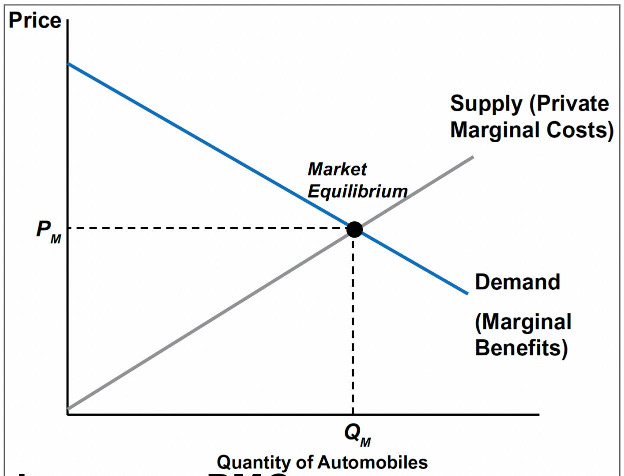
# Externalities

## What is an Externality?

- An **externality** is a side effect of a market transaction—positive or negative—that affects the **well-being of people who are not directly involved** as buyers or sellers.
    - **Negative externality**: imposes costs on others (e.g., air pollution from autos).
    - **Positive externality**: confers benefits on others (e.g., tree planting—amenity value, carbon uptake, habitat).
- Comprehensive analysis must account for **all members of society** (and possibly ecosystems), not only buyers and sellers.

> **Tip**
>
> **Key idea:** Markets maximize total surplus **only** when *all* marginal costs and benefits are internal to decision-makers.

# The Market for Automobiles




- **Supply curve ≈ PMC**: private marginal cost of producing one more unit (labor, materials, energy, etc.).
- **Demand curve ≈ PMB**: private marginal benefit (consumers' willingness-to-pay for one more unit).
- **Market equilibrium** $(P_M, Q_M)$ is efficient **only if** PMC = SMC and PMB = SMB (i.e., no externalities).

## Accounting for Environmental Costs

- Auto production & use create multiple **negative externalities**: local/regional air pollution, $CO_2$, water contamination, toxic wastes, habitat loss, road salt runoff, congestion.
- These costs **do not appear** in the private supply curve; hence **market overstates net social benefits** at $(P_M, Q_M)$.
- We must **internalize** external costs to evaluate the market's true impact.

## EPA Endangerment Finding on GHG Emissions

- **Proposal Date**: July 29, 2025 — EPA formally proposed rescinding the 2009 Greenhouse Gas (GHG) Endangerment Finding (**US EPA**).
- **Statutory Impact**: Without the Endangerment Finding, EPA would **lose authority under CAA §202(a)** to regulate GHG emissions from motor vehicles (**US EPA**; **Wikipedia**).
  - CAA §202(a) is the EPA's authority to regulate tailpipe pollution from new vehicles when it threatens public health or welfare.
- **Regulatory Effect**: Proposal includes **removal of GHG standards** for light-, medium-, and heavy-duty vehicles and engines (**US EPA**).
- **What Remains**: Standards for **criteria pollutants, toxic emissions, corporate average fuel economy (CAFE), and labeling** will **not be changed**(**US EPA**).

## EPA Endangerment Finding: Process & Reactions

- **Public Hearing & Comment Period**:
  - Public hearings held August 19–21 (and possibly August 22), 2025.
  - Comment deadline extended from September 15 to **September 22, 2025** (**Federal Register**; **Office of Advocacy**; **Holland & Knight**).
- **Deregulatory Framing**:
  - The proposal is presented as part of a broader **deregulatory agenda**, hailed as one of the **largest in U.S. history** (**US EPA**).
  - EPA claims the repeal could save Americans **$54 billion annually** and **$170 billion for small businesses** (**US EPA**).
- **Criticisms**:
  - Critics argue the rule **ignores scientific consensus** and is legally weak, potentially dismantling key rules and safeguards designed to protect against climate change (**The Washington Post**).

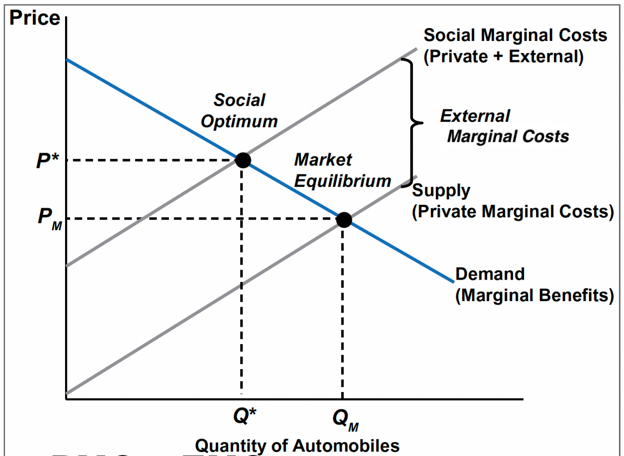
## Valuing Damages

- Assign **monetary values** to external harms to build **SMC**:
  - Direct costs (e.g., **water treatment** for contaminated sources).
  - **Health damages** (medical expenses, morbidity, mortality risk).
  - **Aesthetic/amenity** losses (visibility, recreation quality).
  - **Ecosystem** impacts (biodiversity, habitat quality).
- If we assign no value, the market implicitly assigns **zero**—biasing decisions.

**Warning**

Valuation aggregates **tangible** and **intangible** impacts; precision is hard, but **order-of-magnitude** estimates guide better policy than zeros.

# The Market for Automobiles with Negative Externalities



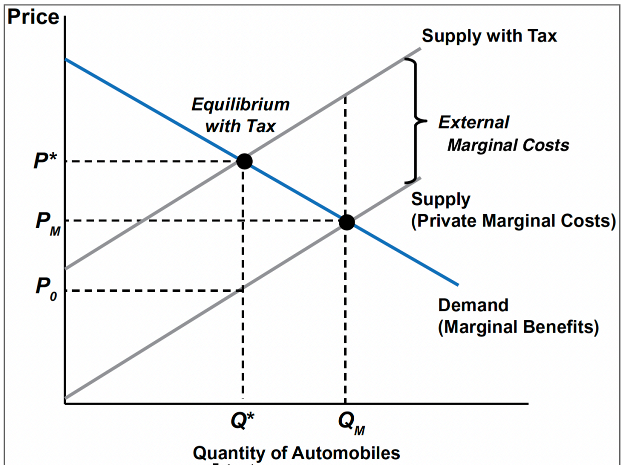

- **SMC = PMC + EMC**, where **EMC** is the *external marginal cost*.
- Typically, **EMC rises** with output when pollution/congestion intensify.
- **SMC** lies **above PMC** by the vertical distance equal to EMC.

# Internalizing Environmental Costs

## Fixing the Inefficiency: Pigovian Tax on Automobiles

- **Market failure:** Prices omit pollution damages ⇒ production **beyond** $Q^*$ is socially undesirable.
- **Goal:** Get the price **“right”** by **internalizing** external costs so they enter consumer/producer decisions.
- **Tool — Pigovian tax:** A **per-unit tax (t)** on automobile **production or emissions** that makes
  $$\mathrm{PMC} + t = \mathrm{SMC} \quad \text{at } Q^*.$$
  - Reflects the **Polluter Pays Principle (PPP)**: those causing pollution **pay its external costs**.

# Automobile Market with Pigovian Tax

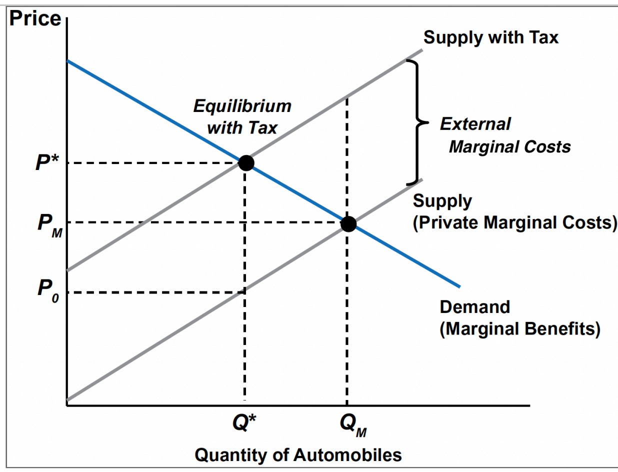



- Impose a **per-unit tax** $t$ equal to **EMC at** $Q^*$.
- Effect: raises firms' **private marginal cost** so the *effective* supply coincides with **SMC**.
- New equilibrium: $(P^*, Q^*)$—the **socially efficient** outcome.
- **Set the right rate**
  - Target the per-unit external damage at $Q^*$ so **PMC +** $t$ **= SMC**.

# Economic Tax Incidence & Elasticities

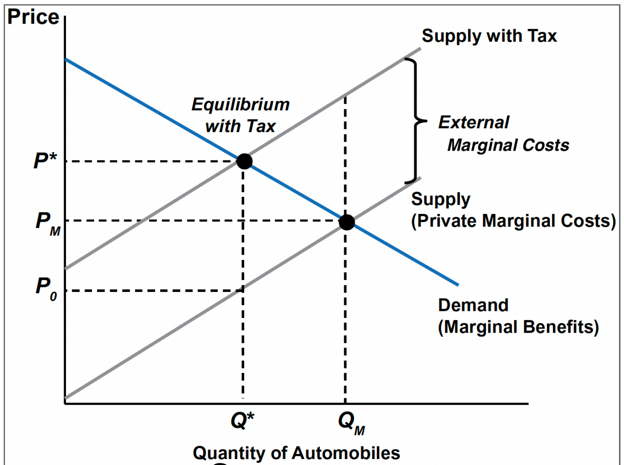
**Note**

- A **price elasticity** of demand/supply measures how responsive the quantity demanded/supplied are to price changes.
- An **inelastic** demand/supply ⇒ quantity changes little for a given price change.
  - The more steeper, the less elastic.




- **Consumer incidence**: the amount of the tax revenue paid by consumers
- **Producer incidence**: the amount of the tax revenue paid by producers

# Who Pays? Economic Incidence & Elasticities




- **Who pays more?**
  - The **less elastic** (steeper) side of the market bears **more** of the tax.
- The claim "**firms pass it all on**" is generally **false**; division is **empirical** and depends on elasticities.

## Distributional Concerns

- **Regressivity:** Energy/fuel taxes can burden lower-income households (higher budget share on energy).
- **How to offset (Revenue recycling):**
  - **Lump-sum dividends** (“carbon cash-back”).
  - **Refundable credits** (e.g., energy affordability credit, EITC (earned income tax credits) top-ups).
  - **Targeted transfers** (rural mileage, extreme-climate heating, heating-oil users).
  - **Fixed bill credits / lifeline blocks** for essential usage *(avoid broad per-unit discounts)*.
  - **Just-transition support** for affected workers/communities.
- **Design notes:** Preserve **price signals** (don’t lower per-unit energy prices), make benefits **visible**, and address **equity/political feasibility**

## Don’t tax everything

- Estimating the “right” tax requires **economics + science**.
- For **small externalities**, the revenue may not justify **measurement + admin** costs.
- Example: “**shirts**” would imply product-by-product rates (cotton vs. synthetics, dye processes, *organic* vs. *recycled*) → **monumental** complexity.

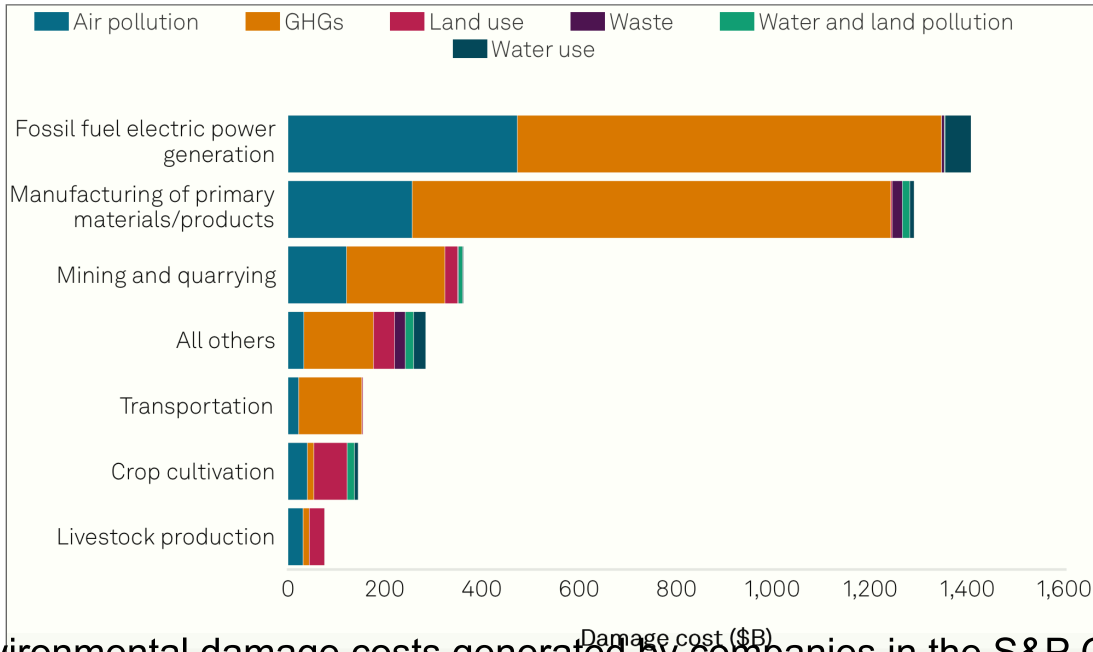
## Go upstream *(tax near resource extraction)*

- Levy at **raw inputs** (e.g., **crude oil**, **cotton**, **ores**, **toxic chemicals**) so costs ripple through prices in proportion to input intensity.
- Focus on materials with **widespread ecological damage** (e.g., **fossil fuels**, key minerals, toxics).
- Benefits: **far fewer tax points**, lower admin burden, avoids bespoke rates on countless final goods.

## Beyond Taxes: Standards & Mandates

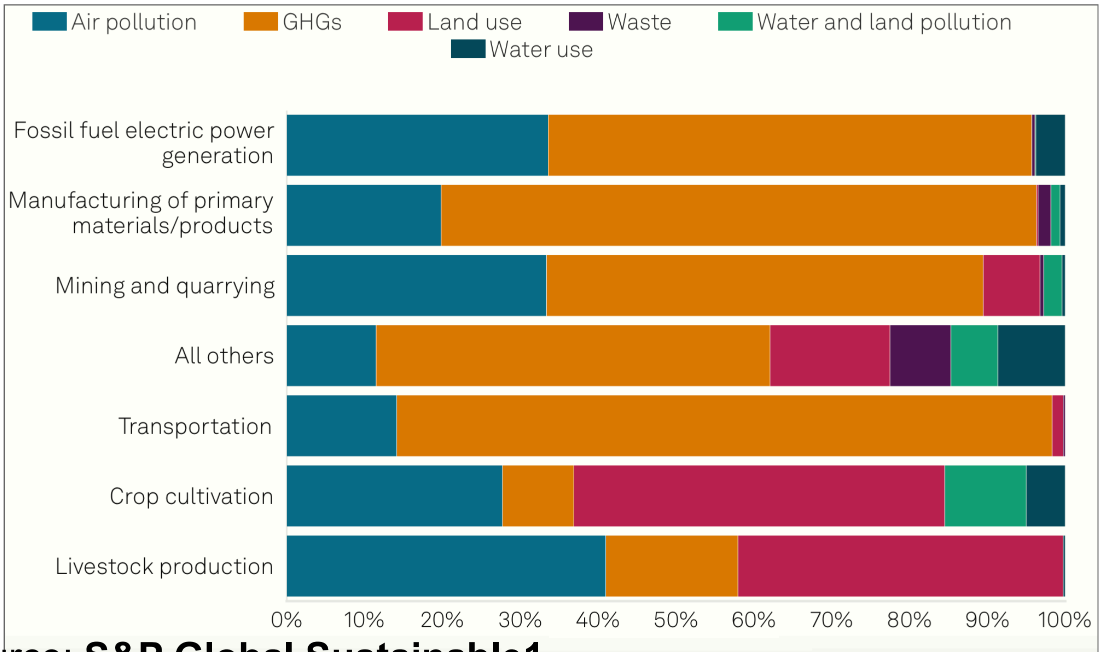
- **Performance/tech standards** (e.g., fuel-efficiency rules) and **mandated controls** (e.g., catalytic converters) can cut emissions **without** directly pricing each unit.
- Standards often raise **upfront/capital costs**, so that prices can rise (**resemble a tax**), while **energy-efficient equipment** uses less fuel/power, so lifetime bills may fall.
- Useful where **damage estimates are uncertain**, **admin is hard**, or **health/ecological thresholds** justify minimum standards alongside prices.

# The Top Externalities of the Global Market




Environmental damage costs generated by companies in the S&P Global Broad Market Index by sector group in 2021.

# The Top Externalities of the Global Market



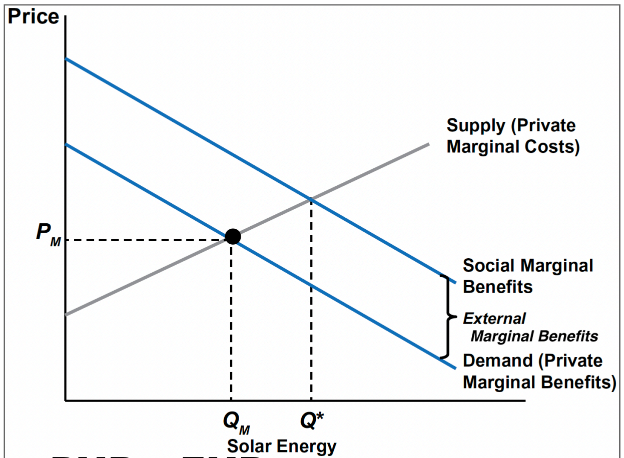

Source: **S&P Global Sustainable1**.

# Positive Externalities

# Positive Externalities: Market Outcome & Policy

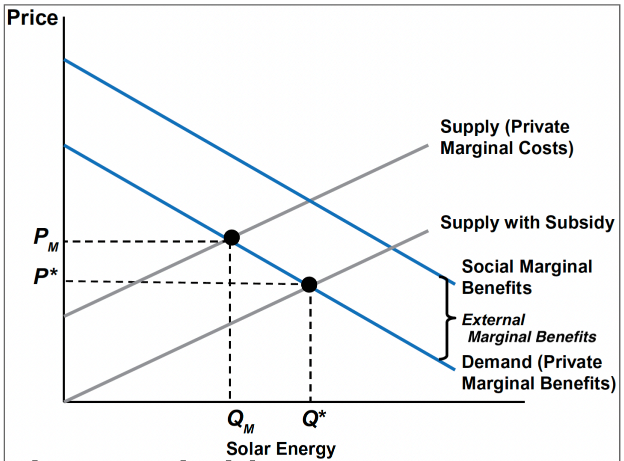
- **Examples**: solar panels (emissions reductions), vaccinations, R&D spillovers, urban trees.
- In an unregulated market, **beneficial spillovers aren't rewarded → underproduction**:
  *PMB* < *SMB* ⇒ $Q_M$ < $Q^*$ → **welfare loss**.
- **Goal:** internalize social benefits to reach the **efficient level** $Q^*$
- **Solar PV example:**
  - Neighbors learn from one household's installation/operation experience.
  - If one home's solar panels make more electricity than that home uses during the day, the extra can flow onto the neighborhood.

# The Market for Solar Energy




- **SMB = PMB + EMB**, where **EMB** is *external marginal benefit*.

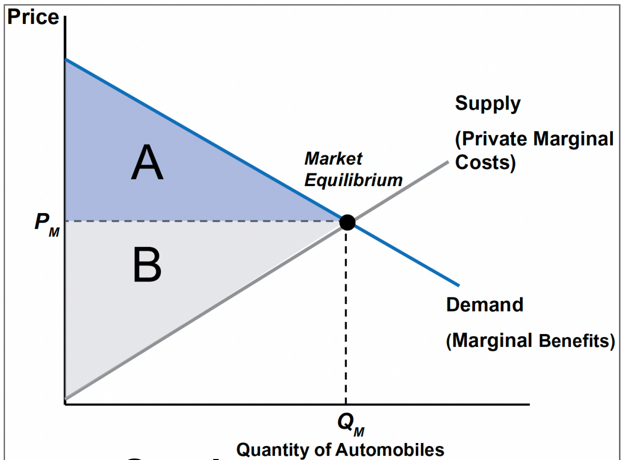
# The Market for Solar Energy with a Subsidy




- A **producer subsidy** of $s$ per unit **lowers** effective marginal cost (downward supply shift).
- A **consumer subsidy (e.g., tax credit)** **raises** effective willingness-to-pay (upward demand shift).
- **Correct subsidy** aligns the market with $Q^*$ where **SMC = SMB**.

# Welfare Analysis of Externalities

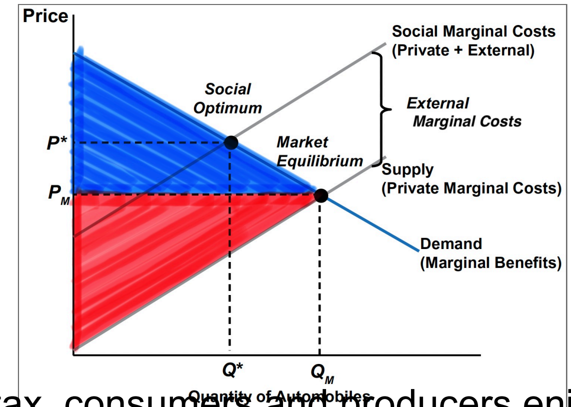
# Welfare Analysis of the Automobile Market




- **Consumer Surplus (CS):** Area **under demand** and **above price**.
- **Producer Surplus (PS):** Area **above supply (PMC)** and **below price**.
- Without externalities, equilibrium maximizes **CS + PS**.

# Welfare Analysis of the Automobile Market

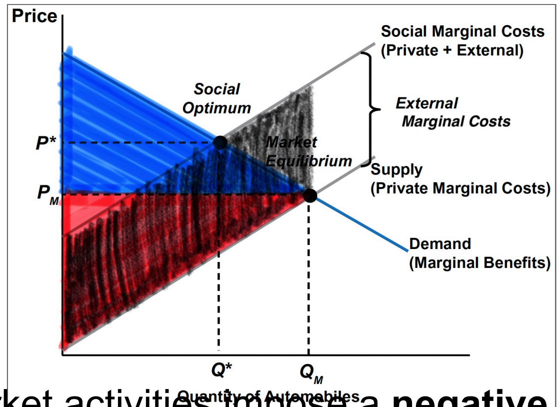
## *Without Pigouvian Tax*




Without Pigouvian tax, consumers and producers enjoys higher **CS** and **PS** than with the tax.

# Welfare Analysis of the Automobile Market

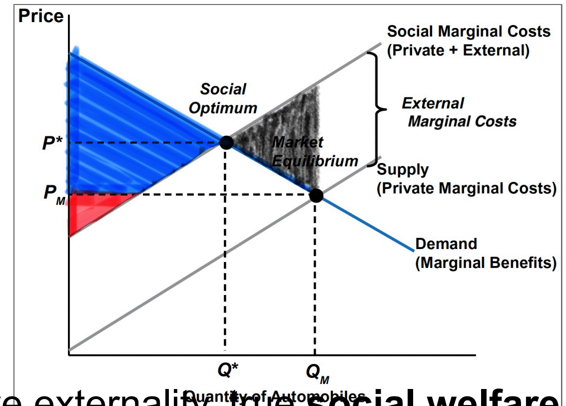
## *Without Pigouvian Tax*




However, these market activities impose a **negative external costs** to third parties, represented by the area **between PMC and SMC** up to the chosen quantity ($Q_M$).

# Welfare Analysis of the Automobile Market

## *Without Pigouvian Tax*

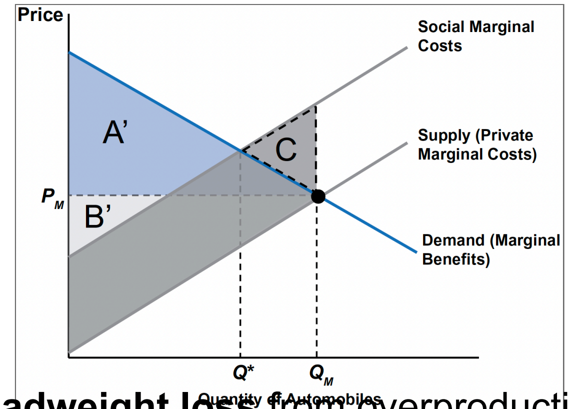



With such a negative externality, true **social welfare (SW)** is:

$$(\text{SW}) = (\text{CS} + \text{PS}) - (\text{External Cost}).$$

# Welfare Analysis of the Automobile Market

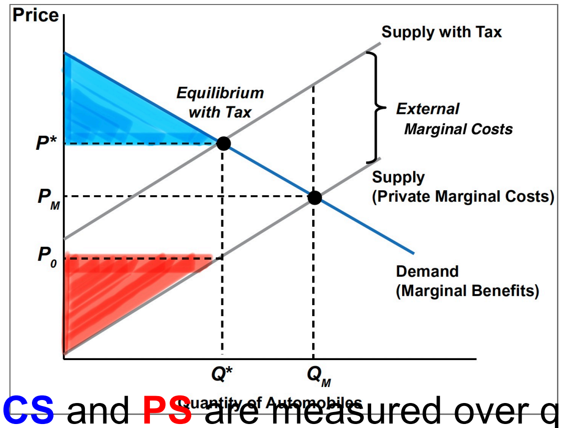
## *Without Pigouvian Tax*




Triangle **C** is the **deadweight loss** from overproduction where **SMC > MB**.

# Welfare Analysis of the Automobile Market

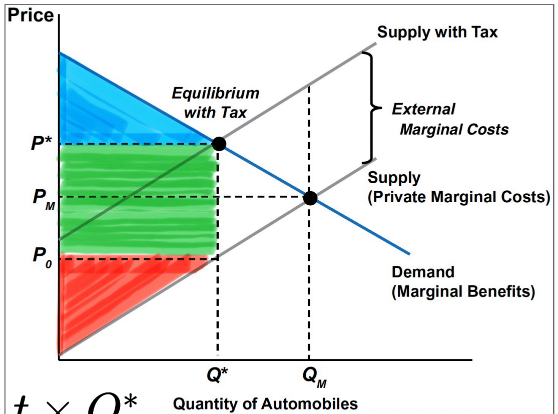
## *With Pigouvian Tax*




With Pigouvian tax, **CS** and **PS** are measured over quantities from 0 to $Q^*$, the new equilibrium quantity.

# Welfare Analysis of the Automobile Market

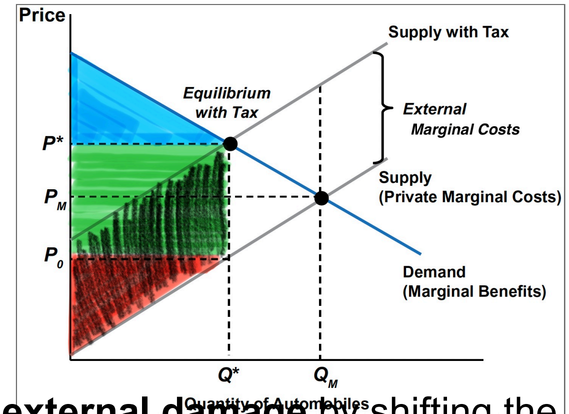
## *With Pigouvian Tax*




**Tax revenue** equals $t \times Q^*$.

# Welfare Analysis of the Automobile Market

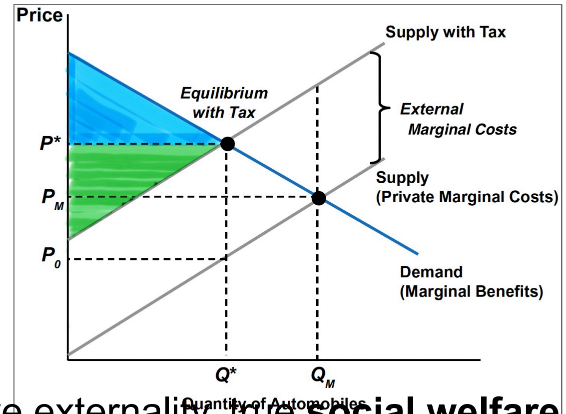
## *With Pigouvian Tax*




Pigouvian tax caps **external damage** by shifting the market to the efficient quantity $Q^*$.

# Welfare Analysis of the Automobile Market

## *With Pigouvian Tax*




With such a negative externality, true **social welfare (SW)** is:

$$(\text{SW}) = (\text{CS} + \text{PS} + \text{Tax Revenue}) - (\text{External Cost}).$$

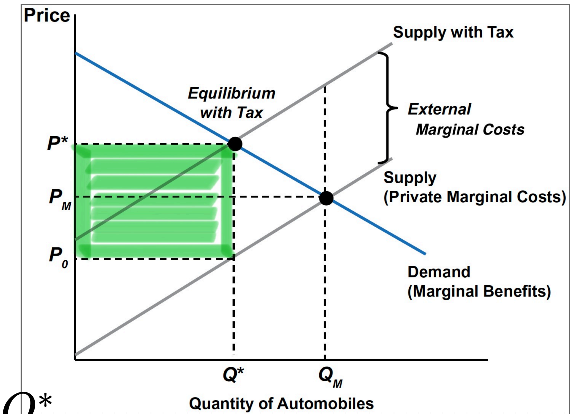
- **Pigovian Tax Improves Welfare**: welfare **rises by C** relative to the unregulated outcome.

## Optimal Pollution

- **Zero pollution** generally implies **zero production**; instead, economists seek the **efficient** (not necessarily zero) level of pollution given current *technology* and *preferences*.
    - The **Pigovian tax** yields **optimal (residual) pollution**—the remaining damages at $Q^*$.
- **Caveat**: If demand shifts right (e.g., autos more valued), the **efficient** pollution level can rise—raising questions about **science-based caps** vs. **pure efficiency**.

> **Note**
>
> - **Preferences** (what people want—demand) and **technology** (what/how much outputs can be produced from scarce inputs—supply), given **endowments** (initial resources) and **prices**, jointly determine **market equilibrium** and **welfare**, as **consumers maximize utility** and **firms maximize profit**.
> - Policy practice often sets **health-based standards** (e.g., the Clean Air Act) and then uses market instruments to **achieve** those standards cost-effectively.

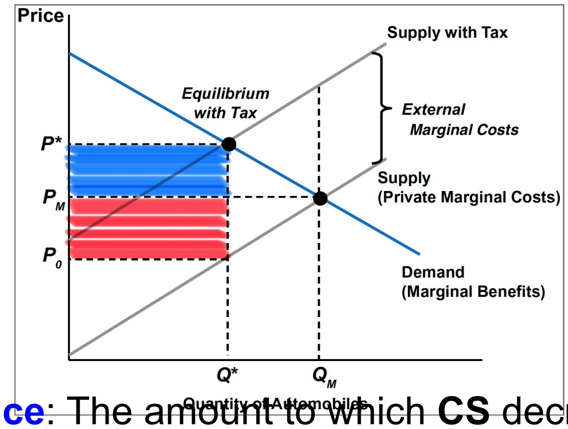
# Tax Incidence




**Tax Revenue**: $t \times Q^*$

**Tax incidence**: The division of the tax burden between consumers and producers, determined by the relative elasticities of demand and supply.

# Tax Incidence




**Consumer incidence**: The amount to which **CS** decreases, measured over quantities from $0$ to $Q^*$ $\rightarrow$ Tax falls on *consumers*, who pay $P^* - P_M$ *more* per unit.

**Producer incidence**: The amount to which **PS** decreases, measured over quantities from $0$ to $Q^*$ $\rightarrow$ Tax falls on *producers*, who receive $P_M - P_0$ *less* per unit.

# Tax Incidence: Ordinary vs. Pigouvian Tax

## Ordinary Tax (in a Competitive Market)

- **Incidence**:
  - Consumers pay higher price ($P^* - P_M$).
  - Producers receive lower price ($P_M - P_0$).
- **Welfare effect**:
  - Creates **deadweight loss (DWL)**.
  - Total surplus falls.

## Pigouvian Tax

- **Incidence**:
  - Consumers and Producers still share burden depending on elasticities.
- **Welfare effect**:
  - **Internalizes externality** (e.g., pollution).
  - Reduces overproduction: $Q_M \to Q^*$.
  - Improves **social welfare** by offsetting external damages.